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Analysis of the dynamic characteristics of a three phase panel type electrostatic actuator using PEM system identification method

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ABSTRACT

This paper describes dynamic characteristics and system identification of a panel type electrostatic actuator which is used for transportation particles and dielectric sheets. The actuator is composed of parallel electrodes, an insulating thin cover-film, a supporting panel and electrode-attachment to the three phase high voltage signals.

Using a digital storage oscilloscope, this study explores the dynamic characteristic of the actuator using system identification analysis. The frequency response data at a discrete set of frequencies obtained from the electrodes are analyzed using the prediction error minimization method (PEM) system identification to reveal important dynamic characteristics including frequency response and transfer function. It is shown that in addition to capacitance effect, the actuator has inductor effect. Based on the proposed model, the effect of signal profile (sine, square and sawtooth) on current consumption and induction current between electrodes is analyzed.

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1. Introduction

In recent years, electrostatic actuator for transportation particles has been described in various papers and proven in application. This method has the capability of sorting, positioning and exact feeding of powder, especially in micro electro mechanical systems (MEMS).

Transportation of particles by a three phase traveling electric field was first proposed by Masuda et al. in [1]. The discovery has been reported as an "electric curtain"(or Masuda panel). Gan-Mor et al. [2] theoretically studied frequency and phase-lag effects on fine particle transportation for the electric curtain modes. Linear motion of microscopic objects has been done by Fuhr et al. [3]. Devices with various novel transportation and manipulation features, utilizing multiphase AC electric fields that instantly generate particle driving forces; have been designed and produced by Moesner et al. [4]. Fantoni et al. [5] described a study of a feeding method based on electrostatic fields. Kawamoto [6] developed five kinds of techniques on electrostatic separation of particle size utilizing the balance of the electrostatic and gravitational force. Also, Kawamoto [7] did numerical and experimental investigations on transport of particles in an electrostatic travelling field. Modeling and parameter identification for a high-power electrostatic linear motor have been done by Yamamoto et al. [8]. In later study, as an electrical model, a six-terminal capacitance network model has been described.

System identification using frequency-domain data is an important tool for prediction dynamic behavior of actuators. Especially in areas where experimental data of the process can be taken relatively cheaply, excitation of the process with periodic signals is an attractive way of extracting accurate information of the system dynamics from experiments. System identification methods can be broadly classified into parametric model-based and non-parametric methods. Parametric model-based system identification methods involve the use of mathematical models to

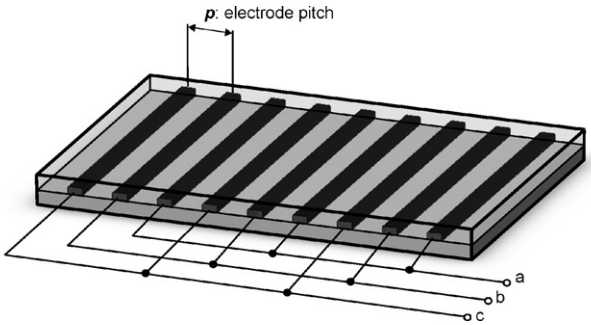


Fig. 1. Basic structure of the *electric panel* with three phase electrode-attachments.

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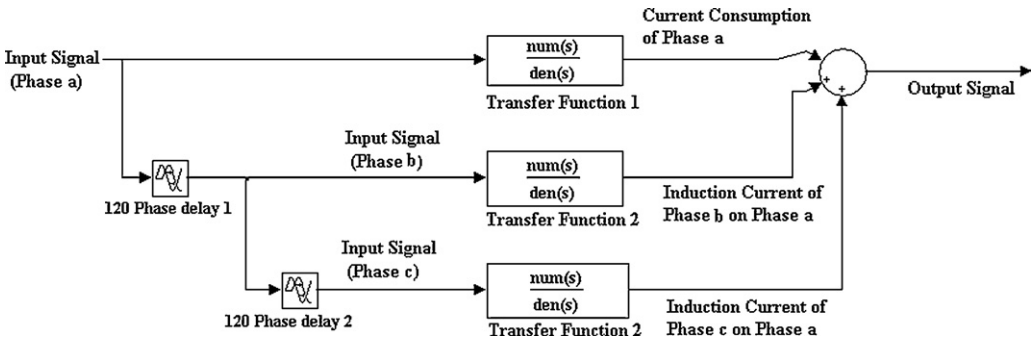


Fig. 2. The schematic diagram of the system.

represent system behavior in either time or frequency-domain. Comparatively, parametric models are more preferred than non-parametric ones. Popular frequency-domain parametric models used for system identification purposes include: state-space models, auto-regressive exogenous (ARX) models, auto-regressive moving average exogenous (ARMAX) models, etc. Many identification algorithms are available to estimate the parameters of such parametric models, e.g., prediction error minimization method (PEM), least squares estimation (LSE), instrumental variable method (IV), maximum likelihood algorithm (MLA), subspace methods [9,10].

In this research, focus is made to system identification and investigation of dynamic characteristics of a three phase panel type electrostatic actuator. The design of power supply of a three phase panel type electrostatic actuator should take into consideration dynamic responses of each individual phase. By use of function generator and high voltage amplifier, high voltage sine wave serving as input signal for one phase of the actuator is generated. Out put (current) signals from the actuator in response to the sine voltage wave input are monitored by a digital storage oscilloscope for acquiring the amplitude and phase of the out puts. Then the system identification method is employed to characterize the dynamic characteristics of the actuator. Choosing a reliable system identification model would ensure accurate characterization of a system's dynamic performance. It is shown that measurements of an unknown system can be employed to approximate a linear model that best describes the dynamic behavior of the system. Models whether for simulation, prediction or design are easier to handle if the number of model parameters is finite, and preferably small. In this study, state-space model is employed for system identification. It is because the simulated outputs of state-space model are best fitted with measured out puts. The system identification method involves the use of PEM method to extract system parameters directly from measured data. Furthermore, the simulated and measured results of system are compared to prove the accuracy of model. Finally, the effect of signal profile (sine, square and sawtooth) on current consumption and induction current between electrodes is shown.

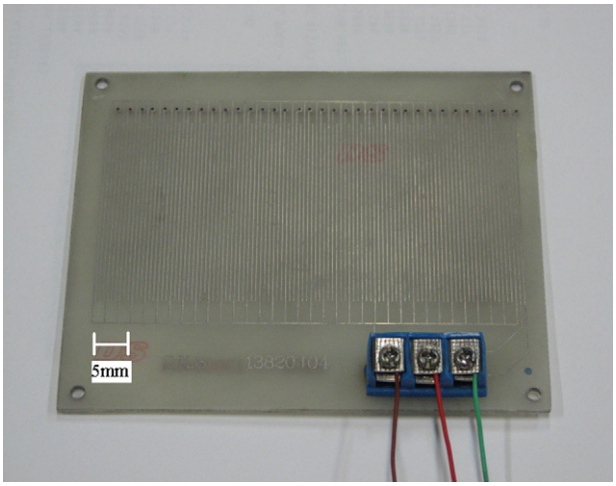


Fig. 3. Prototype of electric panel.

2. Three phase panel type electrostatic actuator for particles transportation

Different actuators for particle handling using an AC electric field wave have been formerly proposed by Moesner et al. [4]. One of such is *electric panel*, as seen in Fig. 1, is composed of parallel electrodes,

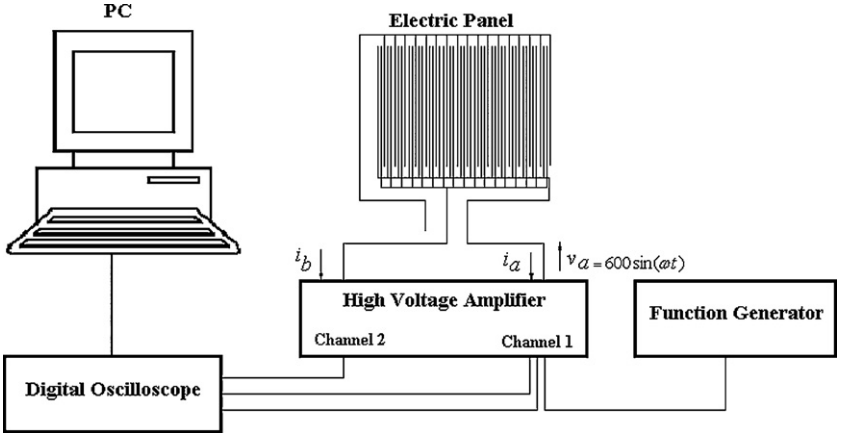


Fig. 4. Signal flow within the measuring system.

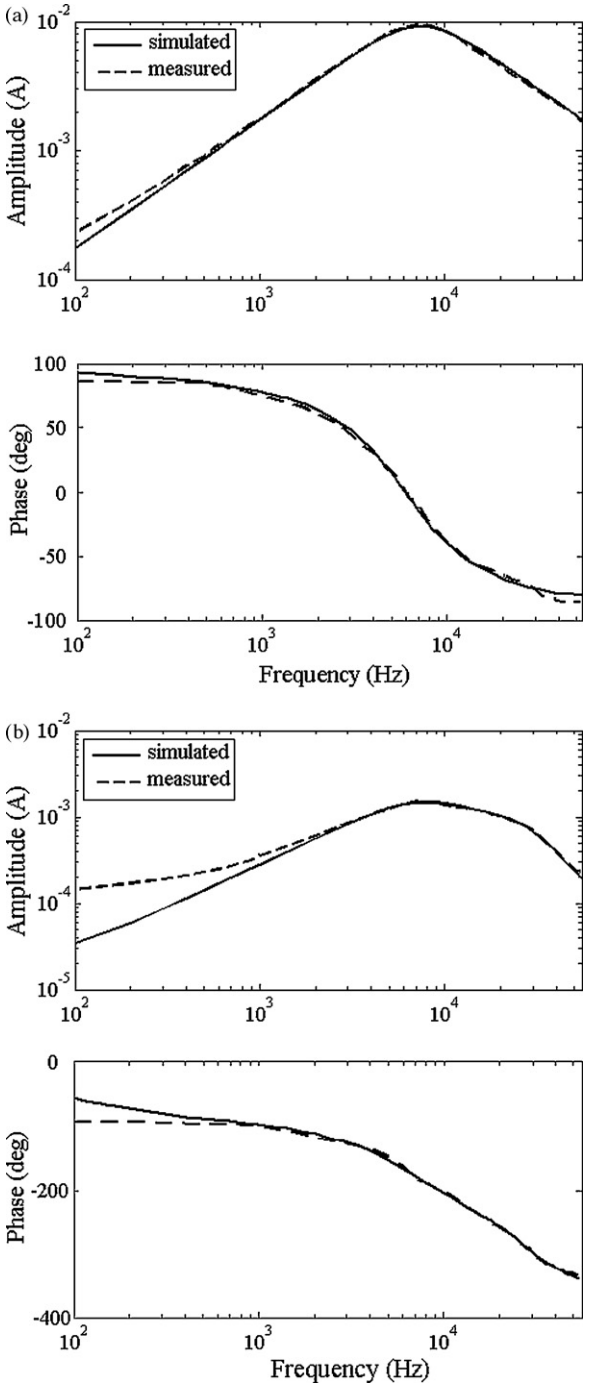


Fig. 5. Comparison between measured and simulated Frequency response of: (a) subsystem-1 and (b) subsystem-2.

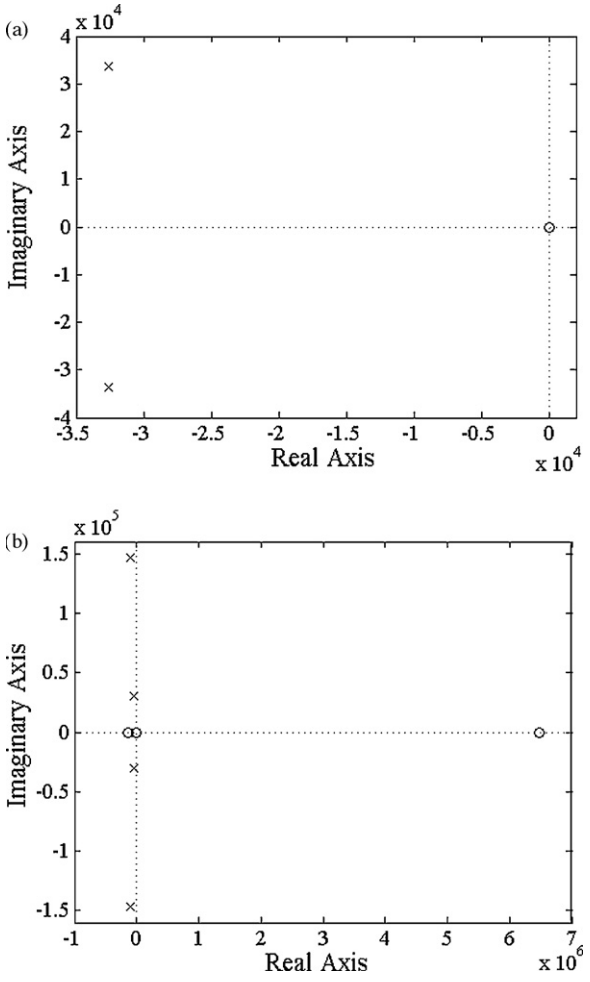


Fig. 6. Pole-zero diagrams of the relative transfer functions of: (a) subsystem-1 and (b) subsystem-2.

an insulating thin cover-film, a supporting panel, and electrode-attachment to the power supply. Balanced, 3-phase high voltage signals (wave) from a programmable source are supplied to the electrodes where every third electrode belongs to the same phase state: a, b and c. Upon the activation of the electrodes, a traveling electric field is created around the electrodes, transiently varying in sync with the applied voltage phase. Particles on the thin insulating film above the parallel field electrodes become charged through the application of voltages to the electrodes. Dynamic forces of this traveling electric field wave overcome adhesion and gravitational forces acting on the charged particles and actuate them in a direction in the plane perpendicular to the electrodes in the electric field traps from electrode to electrode. In this research, electric panel device has been used for experimental investigations. Since the behaviors of phases in the actuator are similar to each other, each phase was considered as a separate system and just one phase selected to do experiments as main system.

3. System identification method

The inverse problem of identifying system parameters from measured dynamic response of systems has attracted a great deal of interest from researchers and several methods have been developed based on the use of state-space models or linear models. Factors like flexible experiment design, ability to obtain physical model and availability of software to perform the estimation are the main reasons for its increasing popularity.

The input-output behavior of the system at any frequency can be expressed by two parameters- Gain (M) and phase (ϕ)-representing the amount of attenuation/magnification and phase-shift respectively. Both of these parameters are functions of frequency and can be obtain easily from frequency response analysis experiment on the system under test. The frequency response is linked to the Laplace model of the input-output relation by evaluation of the transfer function on the imaginary axis. The transfer function of model from the frequency response measured at discrete set of frequencies can be extracted.

To understand the dynamic characteristics of the electric panel would require its transfer functions to be found. It is known that

the ratio of outputs to input is equal to their transfer functions. In electric panel transfer functions are:

$$\frac{i_a}{v_a}(s) = G_1(s) = \frac{b(s)}{a(s)} \quad (1)$$

$$\frac{i_b}{v_a}(s) = G_2(s) = \frac{d(s)}{c(s)} \quad (2)$$

where i_a and i_b denote output currents, v_a is input signal, G_1 is the transfer function of subsystem-1 and G_2 is the transfer function of subsystem-2. The order of polynomial $a(s)$, $b(s)$, $c(s)$ and $d(s)$ depends on the order of dynamics of the system and can be found using system identification algorithms. Fig. 2 depicts the schematic diagram of the whole system. In this system the induction effect of neighboring phases was considered as two disturbances for another phase.

4. Experimentation

Fig. 3 illustrates a prototype of electric panel. Electric panel is fabricated using *flexible printed circuit* technology. The electrode pitch p of the prototype is 600 μm. The electrode width and thickness are 200 μm and 20 μm respectively.

The creation of single phase voltage profile was set with the design and development of a system incorporating a function generator and a dynamic high voltage amplifier (TREK PZD700, Dual Channel) for wave generation. Sinusoidal signal created by function generator was sent to high voltage amplifiers and finally high voltage wave as input signal was sent to one phase electrode in the electric panel. High voltage amplifier is capable of monitoring input signal to the actuator and output (current) signals of the actuator. Output signals were measured in two phases, one (subsystem-1) is the response (current consumption) of one phase of the actuator to the input signal and the other (subsystem-2) is neighboring phase response (induction current) to the input signal. Fig. 4 shows the signal flow within the system. The experiments were performed with varying frequency of input signal from 0.1 kHz to 1 kHz in 0.1 kHz step size then from 1 kHz to 10 kHz in 1 kHz step size then from 10 kHz to 30 kHz in 2 kHz step size and finally from 30 kHz to 55 kHz in 5 kHz step size. The changes in phase and amplitude of the responses were captured and observed by a digital storage oscilloscope and a PC. During experiments maximum voltage for input sinusoidal signal was 600 V.

5. Results and discussion

Fig. 5a and b (dashed line) shows frequency responses characteristics of two mentioned subsystems, obtained under frequencies of 0.1–55 kHz and input voltage $V_a = 600\sin(\omega t)$. The magnitude response between 7 kHz and 8 kHz shows resonant behavior.

The output signal data obtained from F-domain sine wave response are then quantified by the MATLAB system identification toolbox. The steps involved are as follows.

5.1. Handling data

In the MATLAB system identification toolbox, signals and observed data are represented as column vectors. For frequency-domain data, import data include frequency vector, amplitude vector and phase vector.

5.2. Estimating model

Estimating models from data is the central activity in the system identification toolbox. Parametric methods are more preferred in system identification. In this method, a specific model structure is assumed, and the parameters in this structure are estimated using data. This opens up a large variety of possibilities, corresponding to the different ways of describing the system. The most important model structures include the state-space description, as well as several variants of difference equation descriptions. The

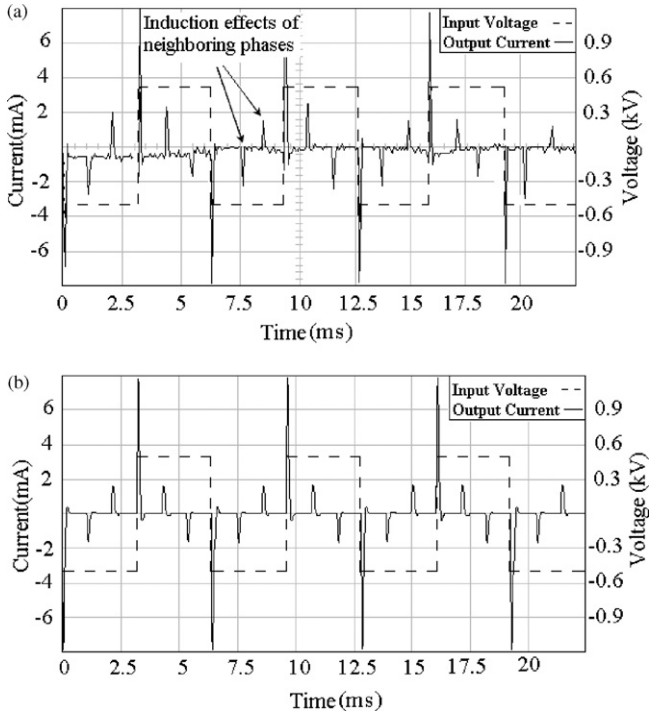


Fig. 7. Comparison between measured (a) and simulated (b) system identification results of electric panel for square signal profile input (V_{max} = 500 V, frequency = 150 Hz).

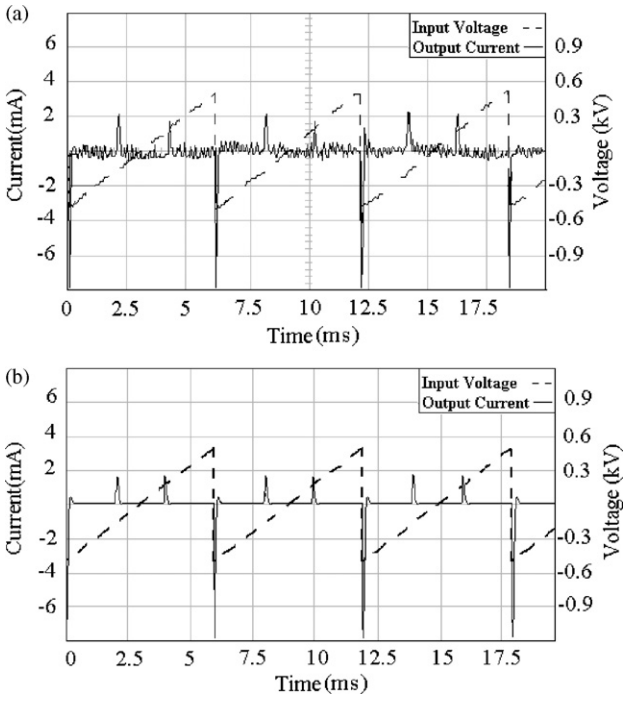


Fig. 8. Comparison between measured (a) and simulated (b) system identification results of electric panel for sawtooth signal profile input (V_{max} = 500 V, frequency = 150 Hz).

models are fed with measured frequency response. The percentage of the output variations that is reproduced by the model is the criterion for fitness. The precise definition of the criterion (FIT) is:

$$\mathrm{FIT} = \left[1 - \frac{\mathrm{NORM}(Y - \mathrm{YHAT})}{\mathrm{NORM}(\mathrm{Y} - \mathrm{MEAN}(\mathrm{Y}))}\right] \times 100 \quad (3)$$

where Y is the measured output and YHAT is the simulated model output. Higher number of FIT means a better model. Among various models, state-space model using PEM identification algorithm acquired best fit in both sub system-1 and sub system-2 with 93.8% and 91.7% of fitness respectively.

Fig. 5a and b (continues line) shows the simulated output of selected models. In this figure the amplitude and phase curves of the measured and the model frequency response are shown. Also, this figure illustrates errors between simulated models and measured data.

5.3. Obtaining relative transfer function

Eqs. (4) and (5) show the obtained transfer functions for subsystem-1 and subsystem-2.

$$G_1(s) = \frac{0.9995s}{s^2 + 65190s + 2.189e9} \quad (4)$$

$$G_2(s) = \frac{0.00507s^3 - 32216.6s^2 - 4.6766e9s + 2.006e12}{s^4 + 2.598e5s^3 + 4.566e10s^2 + 2.46e15s + 6.261e19} \quad (5)$$

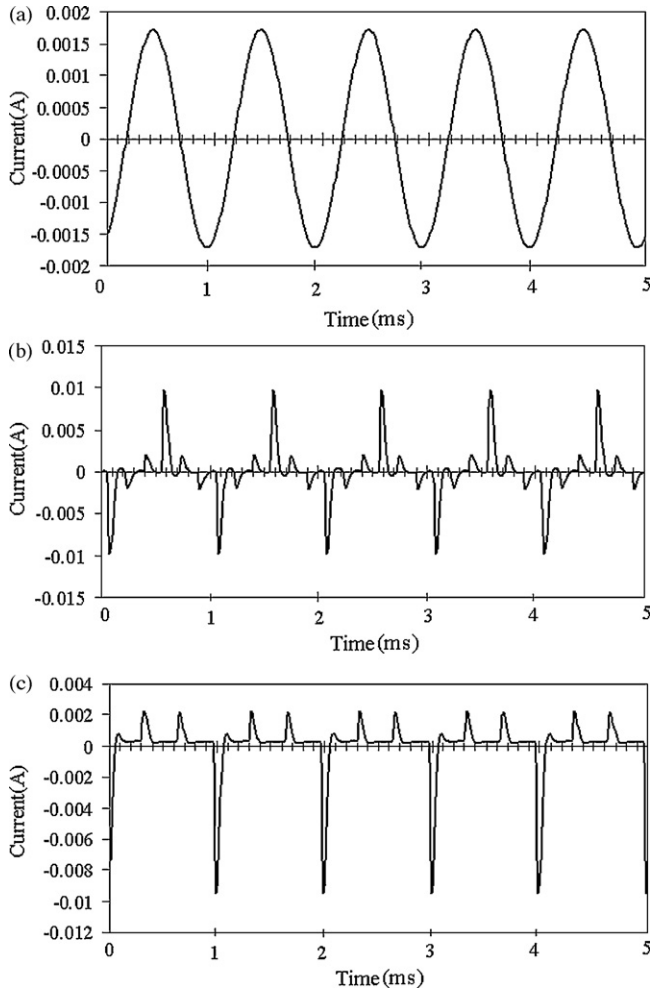


Fig. 9. Predicted current of one phase of electric panel using (a) sinusoidal (b) square (c) sawtooth signal profiles as input signal in frequency of 1000 Hz (V_{max} = 500 V).

Fig. 6(a–b) shows the pole-zero diagrams of the relative transfer functions of the electric panel in S-domain. The order of transfer function (G_1) shows that subsystem-1 is composed of resistance, capacitance and inductor.

Figs. 7 and 8 illustrate comparison between the simulated and measured results of electric panel using square and sawtooth signal profile (V_{max} = 500 V, Frequency = 150 Hz) as input signal respectively. As can be seen, outcomes obtained in simulation by both square and sawtooth signal profile are similar to measured output but some discrepancies in current amplitude exist. Possible reason for the disagreements is the effect of noise in the system.

Figs. 9 and 10 depict predicted current consumption of one phase of electric panel using various signal profiles (V_{max} = 500 V) as input signal in frequencies of 1000 Hz and 5000 Hz respectively. It can be seen from Figs. 9 and 10 that by increasing frequency (up to resonant frequency) impedance decreases and current increases. The increase of current consumption is more significant for sinusoidal wave than square and sawtooth signal profile. It is predictable that the trend will be reversed by increasing frequency more than resonant frequency. Also it can be seen that square signal causes the highest peak in current consumption. By adjusting power supply current limit to be a little higher than predicted peak current, safer power supply operation is achievable.

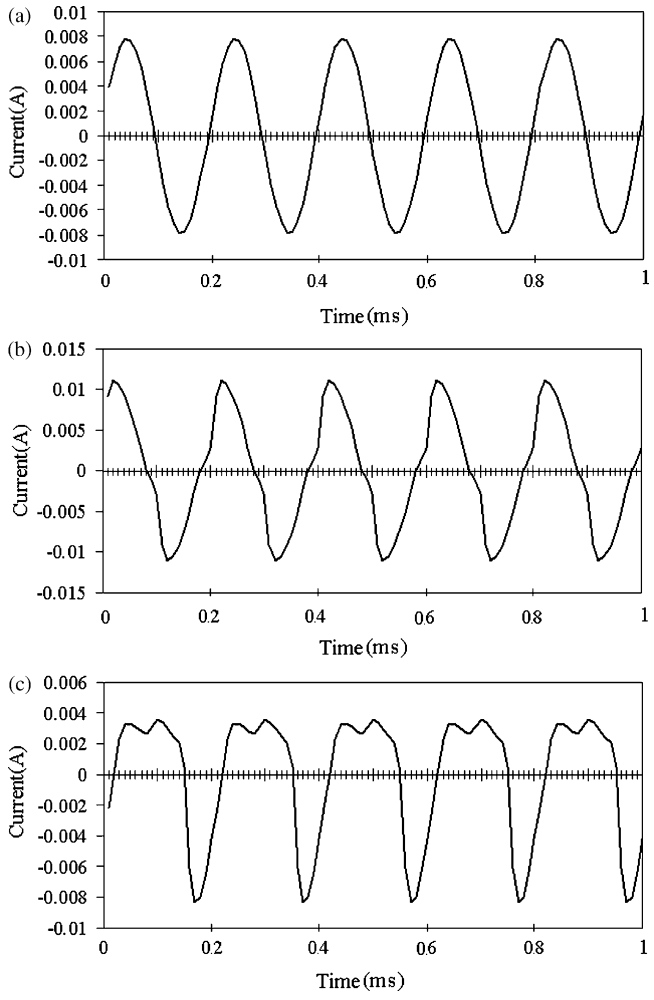


Fig. 10. Predicted current of one phase of electric panel using (a) sinusoidal (b) square (c) sawtooth signal profiles as input signal in frequency of 5000 Hz (V_{max} = 500 V).

6. Conclusion

During this research, dynamic characteristics of a three phase panel type electrostatic actuator (electric panel) have been explored using PEM system identification. It was shown that the system identification method can be used for calculating the transfer function and resonant frequency of electrostatic actuator to design a suitable and safer power supply.

As an electrical model, a model included resistant-capacitance-inductor was presented. The current consumption and induction current characteristics of electric panel can be derived from the model. The model parameters were identified by the proposed method using a digital storage oscilloscope and MATLAB system identification toolbox. In previous research on similar electrostatic actuator, e.g. electrostatic motor, the proposed model consisted of resistance and capacitance. Using system identification method showed that in addition to capacitance and resistance effect, the actuator has inductor effect too.

Based on the measured parameters, the current consumption and induction current characteristics were analyzed and compared using various signal profile. The comparison revealed good agreement, verifying the validity of the proposed model and measurement method. Using the proposed electric panel model and parameters identification method, electric panel can be easily evaluated and analyzed, which will facilitate further study for various practical applications.

References

[1] S. Masuda, K. Fujibayashi, K. Ishida, H. Inaba, Confinement and transportation of charged aerosol clouds via electric curtain, Electrical Engineering in Japan 92 (1) (1972) 41–43.
[2] S. Gan-Mor, S.E. Law, Frequency and phase-Lag effects on transportation of particulates by an AC electric field, IEEE Transactions on Industrial Applications 28 (2) (1992) 317–323.
[3] G. Fuhr, R. Hagedorn, T. Müller, Linear motion of dielectric particles and living cells in microfabricated structures induced by traveling electric fields, in: Proceedings of IEEE Micro Electro Mechanical Systems Workshop, Japan, 1991, pp. 259–264.
[4] F. Moesner, T. Higuchi, Electrostatic devices for particle microhandling, IEEE Transactions on Industry Applications 35 (3) (1999) 530–536.
[5] G. Fantoni, M. Santochi, A modular contactless feeder for microparts, CIRP Annals-Manufacturing Technology 54 (1) (2005) 23–26.
[6] H. Kawamoto, Some techniques on electrostatic separation of particle size utilizing electrostatic traveling-wave field, Journal of Electrostatics 66 (2008) 220–228.
[7] H. Kawamoto, K. Seki, K. Kuromiya, Mechanism of travelling-wave transport of particles, Journal of Physics D: Applied Physics 39 (2006) 1249–1256.
[8] A. Yamamoto, T. Niino, T. Higuchi, Modeling and identification of an electrostatic motor, Precision Engineering 30 (2006) 104–113.
[9] P.P.J. Bosch, A.C. Klauw, Modeling, Identification and Simulation of Dynamical Systems, CRC Press, Boca Raton, 1994.
[10] L. Lennart, System Identification Theory for the User, Scholar Book, New York, 1988.

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